Ex. No: 02

Image Acquisition using Web Camera

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- CNN using Custom Images
- 19AI413-Digital Image Processing Techincs
- EVEN SEM (Slot: 4Q1-2 & 4N5-2)

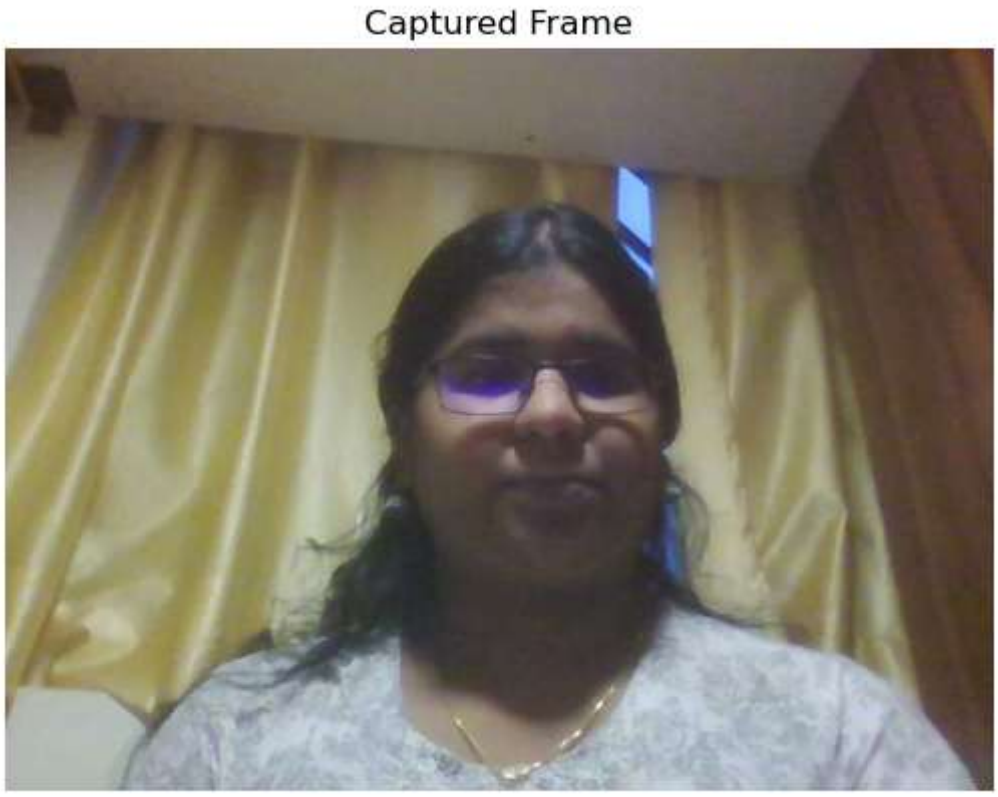
i) Write the frame as JPG file

In [1]:
```
import cv2
import matplotlib.pyplot as plt
from IPython.display import clear_output
import time
```

In [2]:
```
cap = cv2.VideoCapture(0)
ret, frame = cap.read()
if ret:
    cv2.imwrite("captured_frame.jpg", frame)
cap.release()
```

In [3]:
```
captured_image = cv2.imread('captured_frame.jpg')
```

In [4]:
```
plt.imshow(captured_image[:,:,::-1])
plt.title('Captured Frame')
plt.axis('off')
plt.show()
```



ii) Display the video

In [5]:
```
cap = cv2.VideoCapture(0)

for i in range(50):
    ret, frame = cap.read()
    if not ret:
        break
    frame_rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
    clear_output(wait=True)
    plt.imshow(frame_rgb)
    plt.axis('off')
    plt.show()
    time.sleep(0.05)

cap.release()
```



iii) Display the video by resizing the window

In [6]:
```
cap = cv2.VideoCapture(0)

for i in range(50):
    ret, frame = cap.read()
    if not ret:
        break
    resized_frame = cv2.resize(frame, (100, 150))  # Resize to 320x240
    frame_rgb = cv2.cvtColor(resized_frame, cv2.COLOR_BGR2RGB)
    clear_output(wait=True)
    plt.imshow(frame_rgb)
    plt.axis('off')
    plt.show()
    time.sleep(0.05)

cap.release()
```



iv) Rotate and display the video

In [7]:
```
cap = cv2.VideoCapture(0)

for i in range(50):
    ret, frame = cap.read()
    if not ret:
        break
    rotated_frame = cv2.rotate(frame, cv2.ROTATE_90_CLOCKWISE)
    frame_rgb = cv2.cvtColor(rotated_frame, cv2.COLOR_BGR2RGB)
    clear_output(wait=True)
    plt.imshow(frame_rgb)
    plt.axis('off')
    plt.show()
    time.sleep(0.05)

cap.release()
```



In [ ]: